Datathon 2025/26

LSESU Mathematics x LSESU Data Science and Quantitative
Sponsored by Susquehanna International Group (SIG)

Friday 6th March, 2026

1 First, a Message from our Sponsor:

'Susquehanna is a global quantitative trading firm powered by scientific rigor, curiosity, and innovation. Our culture is intellectually driven and highly collaborative, bringing together researchers, engineers, and traders to design and deploy impactful strategies in our systematic trading environment. To meet the unique challenges of global markets, Susquehanna applies machine learning and advanced quantitative research to vast datasets in order to uncover actionable insights and build effective strategies. By uniting deep market expertise with cutting-edge technology, we excel in solving complex problems and pushing boundaries together.'

2 Key Times

- The start date and time is ***Friday 6th March at 5:30pm***.
- You have ***72 hours*** to write and finalise your report and code.
- The end date and time is ***Monday 9th March at 5:30pm***.

3 'What Should I Submit?'

Submissions ***must*** be sent to the following email address by ***exactly one*** member of your team, ***before*** the end of the competition: *mathematics@lsesu.org*. You should also cc in *datascience@lsesu.org* with the email header **Datathon 2025/26**. You should submit:

- A report investigating the direction of your team's chosen analysis. There is a page limit on the report: 10 pages of A4 **MAX**. Only include the key graphs that are essential to your analysis. Other figures can be left in the Jupyter Notebook. You should also provide a full bibliography, which is not included in the page count. Submit as a PDF file.
- A commented Jupyter Notebook file containing the code for other visualisations, and links to any external datasets used.

4 Problem Statement

4.1 Objective

You are providing an advisory service to TFL, who are working on a potential expansion to the system of London Tube stations. They are looking to build a new station, and they would like you to decide upon where it should be.

4.2 Assumptions

- Suppose that TFL has the administrative power to completely remove buildings and use that space.
- Assume that passenger behaviour (e.g. route choice) adjusts rationally in response to the new station - i.e. people will use it if it meaningfully reduces their journey time.
- All regions of land reserved for buildings (not including roads, pavements etc.) are possible to build upon. Ignore planning permission, heritage site protections, and environmental impact assessments, unless you choose to model them as constraints.
- Assume that bus and cycling routes feeding into the new station can be redesigned to support it.

4.3 Suggested Challenges

- **Location:** Give location details for your proposed location, in the form of coordinates.
- **Routes:** Tube lines passing through the station should be decided upon.
- **Interesting Questions:** Can you propose a timetable for the new station? What is the expected ridership? How does the new station affect crowding at nearby stations? How does your proposed station serve communities of varying levels of prosperity? Are there local bus stations, or other transport links that you could replace with one at your station?

4.4 Suggested Datasets

You are allowed to use any source of data, as long as it is properly cited and linked. The main source of data is the TFL Open Data, which is accessible here. Some particularly useful collections are below.

- TFL's *crowding data* is a collection of datasets demonstrating TFL's rail demand over a period of multiple years. Find some more information about the collection here, and the link to the collection here.
- A collection of various London datasets are provided, including information not only regarding TFL, but business, crime, planning and environmental data here.
- The ONS website hosts many UK datasets, have a look here.
- The TFL GIS Open Data Hub contains datasets regarding the location of public transport stations. See the web app here.

4.5 Evaluation Criteria

- **Methodology and Approach:** Includes data preprocessing, techniques used for modelling, and the rigour of your analytical framework.
- **Visualisation:** Clarity, creativity, and effectiveness of charts, maps, and other visual elements.
- **Innovation:** Originality of your approach, use of novel data sources, and creative problem-solving.
- **General Writing Technique:** Structure, coherence, and overall quality of the written report.